Q1.

Petroxide Industries produces a chemical used in the production of mobile phone covers for a mobile phone company.

The chemical becomes less effective when the mean level of impurity is greater than 3 per cent.
Sunita is the Quality Control manager at Petroxide Industries. After a complaint from the mobile phone company, Sunita obtains a random sample of this chemical from 9 batches. She measures the level of impurity, X per cent, in each sample.

The summarised results are as follows.

$$\sum x = 28.8 \qquad \sum (x - \bar{x})^2 = 0.6$$

(a) (i) Investigate using the 5% level of significance whether the mean level of impurity in the chemical is greater than 3 per cent.

(7)

(ii) State the assumption that it was necessary for you to make in order for the test in part **(a)(i)** to be valid.

(1)

(b) State the changes that would be required to your test in part **(a)** if you were told that the standard deviation of the level of impurity is known to be 0.25 per cent.

(2)

(Total 10 marks)

Q2.

A supermarket buys pears from a local supplier. The supermarket requires the mean weight of the pears to be at least 175 grams. William, the fresh-produce manager at the supermarket, suspects that the latest batch of pears delivered does not meet this requirement.

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(a) William weighs a random sample of 6 pears, obtaining the following weights, in grams.

160.6 155.4 181.3 176.2 162.3 172.8

Previous batches of pears have had weights that could be modelled by a normal distribution with standard deviation 9.4 grams. Assuming that this still applies, show that a hypothesis test at the 5% level of significance supports William's suspicion.

(7)

(b) William then weighs a random sample of 20 pears. The mean of this sample is 169.4 grams and s = 11.2 grams, where s^2 is an unbiased estimate of the population variance.

Assuming that the population from which this sample is taken has a normal distribution but with unknown standard deviation, test William's suspicion at the **1%** level of significance.

(5)

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(c) Give a reason why the probability of a Type I error occurring was smaller when conducting the test in part (b) than when conducting the test in part (a).

(1)

(Total 13 marks)

Q3.

Lorraine bought a new golf club. She then practised with this club by using it to hit golf balls on a golf range.

After several such practice sessions, she believed that there had been no change from 190 metres in the mean distance that she had achieved when using her old club.

To investigate this belief, she measured, at her next practice session, the distance, x metres, of each of a random sample of 10 shots with her new club. Her results gave

$$\sum x = 1840 \qquad \text{and} \qquad \sum (x - \bar{x})^2 = 1240$$

Investigate Lorraine's belief at the 2% level of significance, stating any assumption that you make.

(Total 7 marks)

Q4.

Judith, the village postmistress, believes that, since moving the post office counter into the local pharmacy, the mean daily number of customers that she serves has increased from 79.

In order to investigate her belief, she counts the number of customers that she serves on 12 randomly selected days, with the following results.

88 81 84 89 90 77 72 80 82 81 75 85

Stating a necessary distributional assumption, test Judith's belief at the 5% level of significance.

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(Total 9 marks)

Q5.

Bishen believes that the mean weight of boxes of black peppercorns is 45 grams. Abi, thinking that this is not the case, weighs, in grams, a random sample of 8 boxes of black peppercorns, with the following results.

44 44 43 46 42 40 43 46

(a) (i) Construct a 95% confidence interval for the mean weight of boxes of black peppercorns, stating any assumption that you make.

(6)

(ii) Comment on Bishen's belief.

(2)

(b) (i) Abi claims that the mean weight of boxes of black peppercorns is less than 45 grams. Test this claim at the 5% level of significance.

(6)

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(ii) If Bishen's belief **is** true, state, with a reason, what type of error, if any, may have occurred when conclusions to the test in part (b)(i) were drawn.

(2)

(Total 16 marks)

Q6.

The management of the Wellfit gym claims that the mean cholesterol level of those members who have held membership of the gym for more than one year is 3.8.

A local doctor believes that the management's claim is too low and investigates by measuring the cholesterol levels of a random sample of 7 such members of the Wellfit gym, with the following results:

4.2 4.3 3.9 3.8 3.6 4.8 4.1

Is there evidence, at the 5% level of significance, to justify the doctor's belief that the mean cholesterol level is greater than the management's claim? State any assumption that you make.

(Total 8 marks)

Q7.

Jasmine's French teacher states that a homework assignment should take, on average, 30 minutes to complete.

Jasmine believes that he is understating the mean time that the assignment takes to complete and so decides to investigate. She records the times, in minutes, that it takes for a random sample of 10 students to complete the French assignment, with the following results:

29 33 36 42 30 28 31 34 37 35

(a) Test, at the 1% level of significance, Jasmine's belief that her French teacher has understated the mean time that it should take to complete the homework assignment.

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(7)

(b) State an assumption that you must make in order for the test used in part (a) to be valid.

(1)

(Total 8 marks)

Q8.

A jam producer claims that the mean weight of jam in a jar is 230 grams.

(a) A random sample of 8 jars is selected and the weight of jam in each jar is determined. The results, in grams, are

220 228 232 219 221 223 230 229

Assuming that the weight of jam in a jar is normally distributed, test, at the 5% level of significance, the jam producer's claim.

(9)

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(b) It is later discovered that the mean weight of jam in a jar is indeed 230 grams.

Indicate whether a Type I error, a Type II error or neither has occurred in carrying out the hypothesis test in part (a). Give a reason for your answer.

(2)

(Total 11 marks)

Q9.

Bottles of sherry nominally contain 1000 millilitres. After the introduction of a new method of filling the bottles, there is a suspicion that the mean volume of sherry in a bottle has changed.

In order to investigate this suspicion, a random sample of 12 bottles of sherry is taken and the volume of sherry in each bottle is measured.

The volumes, in millilitres, of sherry in these bottles are found to be

996	1006	1009	999	1007	1003
998	1010	997	996	1008	1007

Assuming that the volume of sherry in a bottle is normally distributed, investigate, at the 5% level of significance, whether the mean volume of sherry in a bottle differs from 1000 millilitres.

(Total 10 marks)

Q10.

The lifetime, X hours, of Everwhite camera batteries is normally distributed.
The manufacturer claims that the mean lifetime of these batteries is 100 hours.

(a) The members of a photography club suspect that the batteries do not last as long as is claimed by the manufacturer. In order to investigate their suspicion, the members test a random sample of five of these batteries and find the lifetimes, in hours, to be as follows:

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85 92 100 95 99

Test the members' suspicion at the 5% level of significance.

(9)

(b) The manufacturer, believing that the mean lifetime of these batteries has not changed from 100 hours, decides to determine the lifetime, x hours, of each of a random sample of 80 Everwhite camera batteries. The manufacturer obtains the following results, where x denotes the sample mean:

$$\sum x = 8080 \quad \text{and} \quad \sum (x - \bar{x})^2 = 6399$$

Test the manufacturer's belief at the 5% level of significance.

(8)

(Total 17 marks)

Q11.

The contents, in millilitres, of cartons of milk produced at Kream Dairies, can be modelled

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by a normal distribution with mean 568 and variance σ^2.

After receiving several complaints from their customers who thought that the average content of the cartons had been reduced, the production manager of Kream Dairies decided to investigate.

A random sample of 8 cartons of milk was taken, revealing the following contents, in millilitres.

560 568 561 562 564 567 565 563

Investigate, at the 1% level of significance, whether the average content of cartons of milk is less than 568 millilitres.

(Total 10 marks)

Q12.

Sharon is considering opening a coffee bar. While preparing her business plan, she estimated that a customer will, on average, occupy a seat in the coffee bar for 18 minutes.

She observed customers in a coffee bar in a similar area and recorded the times, in minutes, for which they occupied a seat. The times recorded were as follows:

37.4 18.2 39.0 12.2 46.9 5.4 69.2 16.3 44.6 24.2 39.8

(a) Using the 5% significance level, test the hypothesis that the mean time a customer occupies a seat is 18 minutes. Assume that the data are a random sample from a normal distribution.

(10)

(b) If, in part (a), you had been asked to examine whether the mean time a customer occupied a seat was less than 18 minutes, state the changes, if any, which would have been necessary to:

(i) the null hypothesis;

(ii) the alternative hypothesis;

(iii) the critical value(s);

(iv) the conclusion. Your conclusion should be stated in words; i.e. a conclusion such as “reject H_0” is not sufficient.

(4)

(c) If, in part (a), you had been asked to examine a claim that the mean time a customer occupied a seat was 18 minutes or less, state the changes, if any, which would have been necessary to:

(i) the null hypothesis;

(ii) the alternative hypothesis;

(iii) the critical value(s);

(iv) the conclusion. Your conclusion should be stated in words; i.e. a conclusion such as ”reject H_0” is not sufficient.

(4)

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(Total 18 marks)

Q13.

A large organisation wishes to improve its accessibility to the public. It aims to reduce the mean telephone waiting time (i.e. the time taken to answer a telephone call and to connect the caller with an appropriate member of the Organisation). A number of test calls were made at random times during working hours and the waiting times, in seconds, were recorded as follows.

27 61 48 40 93 47 69 33 72

Assume that waiting times are normally distributed.

(a) Stating your hypotheses, carry out an appropriate hypothesis test to examine whether the mean waiting time is less than 65 seconds. Use the 1% significance level.

(10)

(b) If in part (a) you had been asked to examine whether the mean waiting time was equal to 65 seconds and to use the 10% significance level, state the changes, if any, that you would have made to:

(i) the null hypothesis;

(ii) the alternative hypothesis;

(iii) the critical value(s);

(iv) the conclusion.

(4)

(c) Explain, in the context of part (a), the meaning of:

(i) a Type I error;

(ii) a Type II error.

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(3)

(Total 17 marks)

Q14.

A railway company claims that its trains arrive at their destinations, on average, not more than 10 minutes after the advertised times.

(a) A random sample of eight trains arrived the following numbers of minutes after the advertised times:

24.5 2.9 17.4 13.0 28.7 13.3 5.0 9.6

*(24.5 represents 24 and a half minutes and **not** 24 minutes 5 seconds.)*

Assuming a normal distribution, test, using the 5% significance level, the railway company's claim.

(10)

(b) A newspaper asks the company to provide further information. The company

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produces records showing that a sample of 190 trains arrived at their destinations a mean of 11.21 minutes after the advertised times, with a standard deviation of 6.93 minutes.

Using this second sample of trains, test, using the 5% significance level, the hypothesis that the company's trains arrive a mean of 10 minutes after the advertised times.

(6)

(c) State, giving a brief explanation in each case, how your conclusion in part (b) would be affected if further investigation showed that:

(i) the sample was random but the distribution was not normal;

(ii) the distribution was normal but the sample was not random.

(4)
(Total 20 marks)

Q15.

An instrument for measuring the speed of passing motorists is tested by a police force. A car is driven at known speeds down a straight road. The error (speed recorded by the instrument minus speed of the car) is observed on eight occasions with the following results in m/s:

4.2 –2.8 3.7 –5.9 0.2 6.4 4.1 –1.9

(a) Stating your null and alternative hypotheses, investigate, at the 5% significance level, whether the instrument is biased (i.e. whether the mean error differs from zero). Assume that the data may be regarded as a random sample from a normal distribution.

(10)

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(b) Tests on a different design of instrument showed that in 20 trials there was a mean error of –0.25. From past experience it is known that these errors are normally distributed with a standard deviation of 0.38.

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(i) Investigate, using the 1% significance level, whether this second instrument, on average, underestimates the speed of cars.

(6)

(ii) Compare the performance of the two instruments and give a reason why the second instrument might be preferred to the first.

(3)
(Total 19 marks)

Q16.

A publisher produced a quick revision book for A Level Mathematics. To demonstrate its effectiveness, the book was offered free of charge to a class of students, two weeks before they took a mock A Level examination. The marks, in the mock examination, of eight of the students who accepted the offer were as follows:

73 49 64 46 73 55 59 52

The mean mark in the mock examination, based on past experience, was expected to be

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50.

(a) Investigate, at the 5% significance level, whether the mean mark for students who accepted the offer differed from 50. State your null and alternative hypotheses.

(11)

(b) State the assumptions you have made in order to carry out the test in part (a).

(2)

(c) Encouraged by the initial trial the publisher offered the book free of charge to all students in a particular local education authority area who were due to take A Level Mathematics.A sample of 89 students, who accepted the offer and used the book, obtained a mean mark of 50.4 with a standard deviation of 12.6 in the examination. The average mark for all students who took the same A Level Mathematics examination without using the revision book was 48.8.

(i) Stating your null and alternative hypotheses, examine, at the 1% significance level, whether the mean mark for students who used the book was greater than 48.8.Assume that the data satisfies all conditions necessary for carrying out the test.

(8)

(ii) Give a reason (other than the possibility of making a Type 1 error) why, however high the marks obtained by the students who used the book, this further trial cannot prove that using the book is an effective way of improving A Level Mathematics marks.

(2)

(Total 23 marks)

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